Universidad del Pacífico

AI and Economic Modeling

Track B: the model behind my thesis

Economic freedom and firm entry–exit decisions

Gabriel Saco

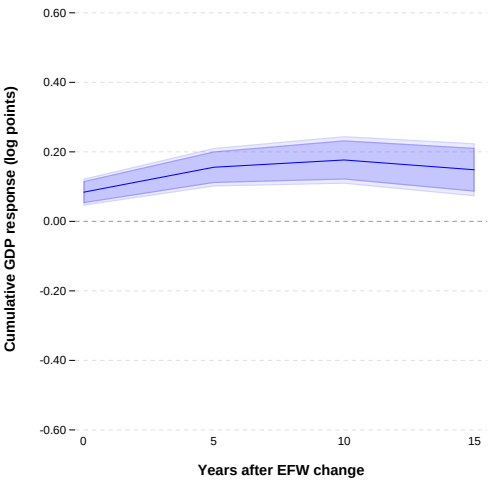
Professor: Alexander Quispe

github.com/gsaco/ai-project

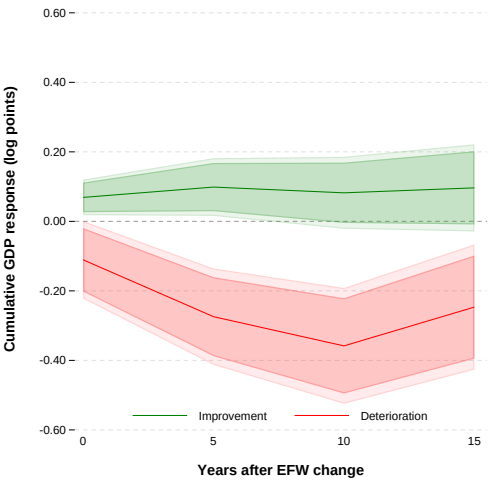
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Why might losses exceed gains?

Symmetry imposed



Asymmetry allowed



From economic freedom to two firm-level channels

Economic freedom concerns the ability to use property, make contracts, and exchange. The Economic Freedom of the World (EFW) index measures several institutional dimensions.

Area 1	Area 2	Area 3	Area 4	Area 5
Size of government	Legal system and property rights	Sound money	Freedom to trade internationally	Regulation

s: Institutionally usable property

Share of existing property available for productive use under prevailing institutional conditions. Title disputes or restrictions on property use can make premises unavailable.

Less usable property: $s \downarrow$

EFW's five areas: Fraser Institute

c: Institutional operating overhead

Recurring institutional expenses paid while operating, independently of current output: permit renewals and fixed costs of reporting or customs administration.

Higher operating overhead: $c \uparrow$

From usable property to production

Fixed property		Usable share		Usable property
$\bar{k} > 0$	$\times$	$s \in (0, 1]$	$=$	$s\bar{k}$
The firm's existing stock.		Available under institutions.		Available for production.

1: Combine property with input

Assume Cobb–Douglas production:

$$q = A(s\bar{k})^{1-\alpha} x^{\alpha}.$$

$x \geq 0$: variable input.
$A > 0$: productivity.
$0 < \alpha < 1$: output elasticity of x.

Output price is one: output equals revenue.

2: Choose convenient units

Measure property in units of the firm's fixed stock: $\bar{k} = 1$.

$$\boxed{q = As^{1-\alpha} x^{\alpha}.}$$

$s = 0.5$: half the existing stock is available for production.

An institutional shock changes the usable share s. The physical stock $\bar{k}$ stays fixed.

One experiment, one changed assumption

Baseline: free restarting

$$\kappa = 0$$

Extension: costly restarting

$$\kappa > 0$$

Institutional path	Shock: $t = 0$	Restored: $t \geq 1$
Deterioration	$s_0 - d$	s_0
No change	s_0	s_0
Improvement	$s_0 + d$	s_0

Same firm and known restoration

Initially open; $0 < d < s_0$, $s_0 + d \leq 1$. Restoration is permanent and known at date 0. Hold $c = c_0$ and other parameters fixed.

What closure means

Free closure preserves property but dismantles the organization. No organization-preserving suspension; financing is available. Discount factor: $0 < \beta < 1$.

The complete firm problem

Activity: $a_t \in \{0, 1\}$

1: open. 0: closed. Initially, $a_{-1} = 1$.

Variable input: $x_t \geq 0$

Choose input when open; set $x_t = 0$ when closed.

Choose activity and input to maximize firm value

$$\max_{\{a_t, x_t\}} \sum_{t=0}^{\infty} \beta^t \left[A s_t^{1-\alpha} x_t^{\alpha} - w x_t - (F + c_t) a_t - \kappa a_t (1 - a_{t-1}) \right].$$

Revenue

$$A s_t^{1-\alpha} x_t^{\alpha}$$

$A > 0$ is productivity. Output price is one.

Input bill

$$w x_t$$

Input quantity times its fixed unit price $w > 0$.

Operating overhead

$$(F + c_t) a_t$$

$F > 0$: ordinary fixed cost. $c_t \geq 0$: institutional overhead. Both paid only when open.

Restart cost

$$\kappa a_t (1 - a_{t-1})$$

Pay $\kappa \geq 0$ only when closed last period and open now. Otherwise, pay zero.

Restart costs leave the input FOC unchanged

1: Hold the operating decision fixed

Choose input x to maximize operating profit:

$$\pi(s,c) = \max_{x \geq 0} \{ A s^{1-\alpha} x^{\alpha} - wx - F - c \}.$$

2: Differentiate with respect to input

The slope of operating profit is:

$$\alpha A s^{1-\alpha} x^{\alpha-1} - w.$$

One extra unit adds revenue and costs w.

3: Set the slope equal to zero

The first-order condition is:

$$\boxed{\alpha A s^{1-\alpha} x^{\alpha-1} = w.}$$

Marginal revenue product = input price.

Diminishing returns give a unique optimum. The FOC is unchanged: rebuilding costs do not depend on x. The changed result concerns closure and recovery.

From optimal input to output and profit

1: Choose input

Rearrange the FOC, then raise both sides to $1/(1-\alpha)$.

$$x^{1-\alpha} = \frac{\alpha A s^{1-\alpha}}{w},$$
$$x^*(s) = s\left(\frac{\alpha A}{w}\right)^{1/(1-\alpha)}.$$

2: Compute output

Substitute optimal input into production.

$$q(s) = sA\left(\frac{\alpha A}{w}\right)^{\alpha/(1-\alpha)}.$$

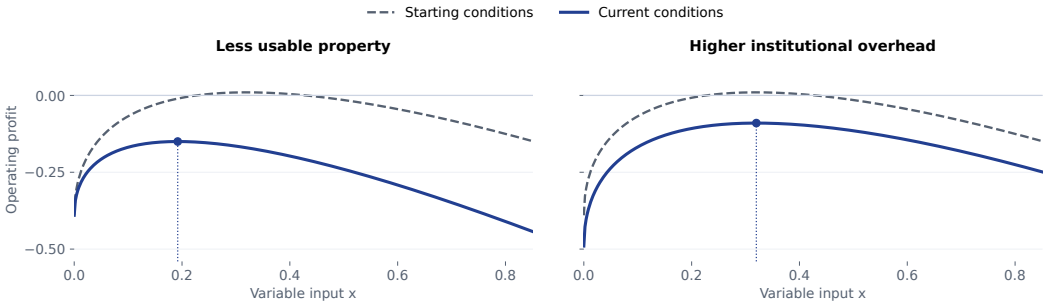
3: Compute operating profit

Multiply the FOC by x^*:
input spending is $wx^* = \alpha q$.

$$\begin{aligned}\pi(s,c) &= q(s) - wx^*(s) - F - c \\ &= (1-\alpha)q(s) - F - c.\end{aligned}$$

$q(s_t)$ is output if operating. Actual output is $y_t = a_t q(s_t)$: $a_t = 1$ if open and 0 if closed.

Two channels, different effects



Conditional on operating	Starting	Lower s	Higher c
Chosen input x^*	0.320	0.192	0.320
Output q	0.80	0.48	0.80
Operating profit π	0.01	-0.15	-0.09

Less usable property changes optimal input and output. Higher overhead shifts profit vertically without changing either operating choice.

Illustrative values: $s : 0.50 \to 0.30$ or $c : 0.09 \to 0.19$, one change at a time.

Closure today and restarting later

Normal operating profit

$$\pi_0 = \pi(s_0, c_0) > 0$$

Deterioration profit

$$\pi_- = \pi(s_0 - d, c_0)$$

Normal business value

$$V = \frac{\pi_0}{1-\beta}, \quad 0 < \beta < 1$$

Keep operating

$$\pi_- + \beta V$$

Current profit plus discounted future earnings.

Close now

$$\beta \max\{V - \kappa, 0\}$$

Closure is free; restarting remains an option.

After institutions are restored

Closure preserves property but dismantles the organization. Rebuilding the workforce and commercial relationships costs κ.

Net value of restarting: $V - \kappa$

$V > \kappa$	$V < \kappa$
Restart	Remain closed

Conjecture: a temporary setback, a lasting loss

Maintain the experiment and assumptions above.

1: Conditions

Improvement profit: $\pi_+ = \pi(s_0 + d, c_0)$

Normal operation pays

$$\pi_0 > 0$$

Positive operating profit under normal conditions.

Continuation loses value

$$\pi_- + \beta V < 0$$

Future earnings cannot offset the current loss.

Rebuilding is too costly

$$\kappa > \pi_+ + \beta V$$

An identical inactive owner would not enter even during improvement.

2: Decisions

Deterioration
Close and never restart.

Improvement or no change
Remain open.

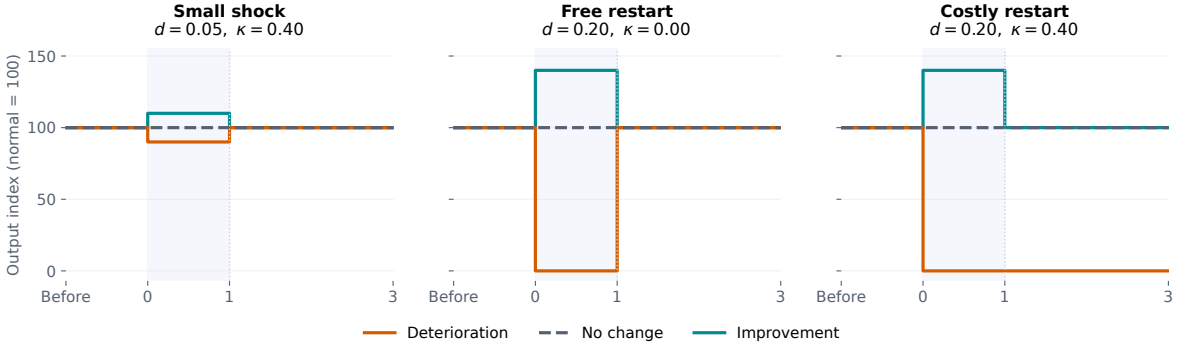
3: Output comparison

Normal output: $q_0 = q(s_0)$.
Loss and gain are relative to no change.

When?	Loss	Gain
During the shock	q_0	$(d/s_0)q_0$
After restoration	q_0	0

Because $d < s_0$, the initial loss exceeds the gain. After restoration, the loss persists while the gain disappears.

Institutions recover. Does production recover too?



Before the shock $\longrightarrow$ Temporary shock: $t = 0$ $\longrightarrow$ Restored: $t \geq 1$

Small shock

Open → open → open.

Free restart

Open → close → reopen.

Costly restart

Open → close → stay closed.

The last two cases use the same shock. Only the restart cost changes.

Prior work and my proposed contribution

Established mechanism

Dixit (1989), pp. 621, 623, 629: organizational decay, temporary reversals, and tolerating losses to avoid rebuilding.

Dixit (1992), pp. 119–120: suspension differs from abandonment. Its appendix (pp. 129–131) sets out the stochastic switching equations.

Roberts and Tybout (1997): evidence of sunk entry costs in exporting; not a test of EFW shocks. Review limited to the author-institution abstract.

Scope of the literature review

Dixit's pp. 625 and 630 refer to appendices A/B available from the author; the published article does not include them.

Princeton paper 91 (1988) is catalogued, with no downloadable text there. Their contents remain outside this review.

I retain costly rebuilding and use known institutional paths to derive the conditional loss–gain comparison. I will then test suspension and entry. This bounded review does not establish originality.

The mechanism and next steps

Institutions

Usable property and overhead change profitability.

Activity

Equal shocks can produce different closure decisions.

Persistence

Costly restarting can prevent output recovery.

Prove by hand

Input optimum; value comparisons for operating, closing, and restarting; conditional output inequalities.

Simulate

Vary margins, shock size and duration, and restart costs. Allow suspension, entry, and heterogeneous firms.

Formalize in Lean

Optimizer and value identities; switching rules and no delayed reentry; conditional inequalities, with logs only for positive output.

The step most likely to fail

A temporary shock may induce **cheap suspension**, without dismantling the organization and requiring costly rebuilding.